



SUPPLY CHAIN INTELLIGENCE // FREE CASE STUDY // LAND DEFENSE // EU (FRANCE/GERMANY)

KNDS

The Maker of the Caesar Howitzer Cannot Make Propellant Fast Enough to Win a War

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KNDS (Krauss-Maffei Wegmann + Nexter) is the Franco-German defence conglomerate that builds the Caesar self-propelled howitzer — the artillery system that has become synonymous with Ukrainian fire superiority. France has delivered 30+ Caesar systems to Ukraine. Germany is the world's largest tank manufacturer. But KNDS systems fire ammunition. And the propellant inside that ammunition — nitrocellulose — is produced by fewer than six qualified European manufacturers, all of them operating at or above capacity, none of them able to scale fast enough to meet a sustained peer-conflict demand. This is the supply chain constraint that limits European artillery effectiveness in any conflict beyond 90 days.

6	30+	90 DAY	ZERO
QUALIFIED EUROPEAN	CAESAR HOWITZERS	ESTIMATED EUROPEAN	NEW EUROPEAN PROPELLANT
NITROCELLULOSE MANUFACTURERS — ALL AT CAPACITY	DELIVERED TO UKRAINE — 6 ROUNDS/MIN	ARTILLERY PROPELLANT RESERVE AT WARTIME	PLANTS COMMISSIONED SINCE 2022

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INTELLIGENCE ALERT // KNDS // THE PROPELLANT TRAP

THE CORE FINDING

KNDS is the most important artillery and armour manufacturer in continental Europe. The Caesar howitzer has proven combat effectiveness in Ukraine. The Leopard 2 is the tank of choice for NATO expansion. The KNDS Main Ground Combat System (MGCS) is the future of European land warfare. All of these systems fire propellant-based ammunition. That propellant — nitrocellulose — is produced by a handful of European chemical manufacturers designed for peacetime consumption rates. Ukraine has demonstrated that modern peer conflict consumes 10–20 times peacetime ammunition consumption. Europe has not built the propellant capacity to sustain that consumption rate. KNDS systems are exquisite. The supply chain behind the rounds they fire is not.

THE NITROCELLULOSE PROBLEM // ARTILLERY WITHOUT PROPELLANT IS SCULPTURE

Nitrocellulose is the primary propellant in all NATO-standard artillery rounds including the 155mm rounds fired by Caesar and the 120mm rounds fired by Leopard 2. It is produced through a dangerous and heavily regulated chemical process — nitration of cellulose — that requires specialised facilities, strict safety controls, and significant capital investment to scale. Unlike semiconductor fabs or shipyards, nitrocellulose plants cannot be rapidly retooled. Building a new qualified propellant plant in Europe takes 5–8 years from planning to production.

MANUFACTURER	COUNTRY	CAPACITY STATUS	VULNERABILITY	KNDS EXPOSURE
Eurencor	France / Sweden / Belgium	Operating above design capacity	France restricted exports to prioritise domestic stock — limits allied resupply	PRIMARY supplier for Caesar rounds — restriction constrains export ammunition
Nitrochemie (RUAG)	Switzerland / Germany	At capacity — 2-year order backlog	Swiss export controls historically neutral — but legally complex for conflict zones	Key supplier for German 155mm and tank rounds
Nobel Sport / Chemring	UK / Norway	Constrained — UK defence priority	UK prioritising own stockpile rebuild — limited availability for continental clients	Partial alternative for KNDS German programmes
Nammo	Norway / Finland	Scaling — 36-month lead for new capacity	NATO-aligned — best positioned for scale-up	Positive — but constrained timeline
MAXAM	Spain / Portugal	Moderate capacity — growing	Spain increasing defence budget — domestic priority	Secondary source — not primary for KNDS core calibres

THE UKRAINE CONSUMPTION SIGNAL // WHAT PEER CONFLICT DOES TO PEACETIME SUPPLY CHAINS

Ukraine has consumed approximately 6,000–10,000 155mm rounds per day during high-intensity operations. Pre-war European production capacity was designed for approximately 300,000–400,000 rounds per year — roughly 800–1,100 per day. Ukraine's consumption rate is 6–12 times the entirety of European peacetime production. European nations have depleted their own stockpiles to supply Ukraine. The reconstruction of those stockpiles — before supplying any future conflict — requires propellant. That propellant does not exist in sufficient quantity.

BOTTOM LINE

KNDS systems are world-class. The Caesar has proven accuracy, mobility, and crew effectiveness in the most demanding combat environment since Korea. The Leopard 2 is the best Western main battle tank in production. None of that matters if the rounds run out. The limiting factor in European artillery capability is not the system. It is the propellant chemistry that feeds it. Defense.Codes tracks this signal weekly.

WHAT A FULL ENGAGEMENT WOULD ADD

A full Defense.Codes engagement would add: (1) Nitrocellulose capacity map — all six qualified European producers, annual throughput, current order backlog, and export restriction status; (2) KNDS programme-by-programme propellant dependency analysis; (3) Wartime consumption model — how long does European propellant stock sustain a Ukraine-intensity conflict before production outpaces supply; (4) Nammo and Eurenco capacity expansion timeline vs demand signal; (5) US Radford Army Ammunition Plant as alternative supply source analysis.

This case study is one of five free public intelligence samples published by Defense.Codes. Category: LAND DEFENSE — EU. Full-depth engagement reports available for purchase. Contact: ceo@capa.cloud | defense.codes

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Sources: KNDS Group corporate publications — Eurenco annual reports — NATO munitions production reports 2024–2025 — IISS Military Balance 2025 — Reuters Ukraine artillery consumption analysis — Nammo capacity expansion announcements — Roland Berger defence supply chain 2025 — European Defence Agency ammunition stockpile reports — Nitrochemie RUAG public disclosures.

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